Harsath Kaliappa Gounder Thangavel

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Education

York University **January 2020 – December 2024**
Bachelor of Science in Computer Science (Honours) *Toronto, Canada*

- Data structures
- Algorithm design
- Data mining
- Database systems
- Computer networking
- Compiler design
- Computer architecture
- Operating systems

Work Experience

Privex **October 2023 – January 2024**
Software Engineer *London, United Kingdom*

- Developed and deployed an AI customer service agent using a self-hosted Large Language Model (LLM) and Retrieval Augmented Generation (RAG) techniques.
- Reduced staff workload by handling approximately 70% of repeated customer inquiries related to technical issues, products, and services.
- Implemented REST APIs in Flask and integrated with Discord API to serve customers on Privex's Discord server.
- Utilized vector databases and MySQL for training data management, allowing dynamic updates from Discord by Privex staff.

CogniPet **April 2019 – January 2020**
Machine Learning Engineer *Zürich, Switzerland*

- Trained and deployed a deep learning model for recognizing 120 breeds of dogs using the Xception neural network with 93% accuracy.
- Implemented REST APIs and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).

Projects

Jix Interpreter | *Interpreter written in C*

- Developed an interpreter for a dynamically typed programming language with support for basic functional programming.
- Implemented features such as fault-tolerant parsing and error handling, dynamic types (int, bool, string, and array), functions, nested blocks, recursion, built-in functions, if-else, return, and break statements, while and for loops.
- Added functional programming support, including first-class functions, higher-order functions, and function pointers.

Cap-Containers | *C header-only containers library*

- Highly-generic, self-contained, header-only containers library written in C.
- Supports sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority), stack, lists, and arena memory allocator.
- Includes thread-safe variants with cross-platform support (Windows, Linux, and macOS).

HashRing | *Go module*

- Consistent hashing data structure implementation in Go using UUIDs and BLOBs.

Blueth | *C++ Linux networking framework*

- Written a networking framework for Linux that provides a set of common client-server abstractions for developing high-performance, concurrent, and event-driven networking services.
- Implemented event-loop abstraction, SOCKS5 client, HTTP parsers and proxies, thread-pool executors, caches, and codecs.

Volunteer Experience

Wikipedia **July 2021 – Present**
Editor

- Contributed over 20,000 edits to Wikipedia, primarily focusing on topics within computer science.
- Elevated 4 data structure and cyber security articles to Good Article status.

Technical Skills

Languages: Python, C++, C, Java, Go, JavaScript, SQL, Perl, Bash, PHP, HTML/CSS, LaTeX
Technologies: Linux/Unix, MySQL/PostgreSQL, Django/Flask, JWTs, CI/CD pipelines, Neo4j
Developer tools: Git, Docker, LXC/LXD, GDB/LLDB, Amazon Web Services